

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE-202 Electrical Power Generation

Date: 28.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Give the classification of Energy sources which are used for Electrical Power Generation. [03]

(b) Draw and explain the Feed water and steam cycle. Clearly define the function of each component. [04]

Q - 2 (a) With neat and clean diagram explain the Coal based steam power plant. [07]

(b) Make the comparative analysis between Nuclear power plant and Steam power plant. [07]

OR

Q - 2 (a) Give the proper placement and working of following component(Diagram necessary): [09]
1. Steam condenser
2. Electrostatic precipitator
3. Dust collector

(b) List out the component of Magneto Hydro Dynamic power plant and explain the function of every component. Also discuss the limitations of this particular power plant. [05]

Q - 3 (a) Give the classification of hydraulic turbine. Briefly Explain any one of them. [06]

(b) Explain with schematic diagram: How the fuel cell based power plant works? [04]

(c) Discuss the major limitations of Nuclear power plant. [04]

OR

Q - 3 (a) Draw the schematic diagram of diesel engine power plant. Also explain the working of essential component. [07]

(b) Draw the layout of Hydro-Electric power plant and discuss the site selection criterion. [07]

SECTION – II

Q - 4 (a) Explain the Modified liquid dominated scheme of Geothermal power plant with necessary diagram. [05]

(b) Write the site selection criterion for Wind power plant. [02]

Q - 5 (a) Draw the neat and clean layout of closed cycle Gas turbine Power plant. Also explain the importance of compressor, starting motor and gas turbine. [07]

(b) Compare the benefits and drawbacks of Horizontal axis and vertical axis wind power plant. [04]

(c) Draw the schematic diagram of wind mill which is generally used for power generation in India. Clearly mention each and every component. [03]

OR

Q - 5 (a) Compare all kind of Solar collectors for the following criteria: Shape, tracking system, efficiency, application. [07]

(b) Draw and explain binary cycle solar thermal power plant. [04]

(c) Write a short note on Utilization of Geo-Thermal energy. [03]

Q - 6 (a) Draw and explain the working of Modified open cycle OTEC plant. [07]

(b) List out the different technique of Tidal energy power plant. Briefly explain any one of them. [07]

OR

Q - 6 (a) Draw and explain the Nuclear reactor. Also give the function of each and every component. [07]

(b) Write a short note on Solar wind hybrid. [04]

(c) Write down the advantages and disadvantages of ocean energy conversation. [03]
